

1 Persona-driven synthesis

Regime A canonical

$N=500$
8 buyer archetypes

Regime B reversals

$N=200$
130 flips,
70 controls

Generation engine

LLMs incl. Qwen-2.5 +
dialogue-state register

700 draft dialogues

fluent but rigid

2 Organic paraphrasing

Voice sampling

5 buyer, 4 merchant voices

Rewrite with hard rules

Gemini Flash-Lite, facts locked

Quality gates

similarity ≤ 0.82 , numbers match

Micro-jitter

lowercase, fillers, sparse typos

700 humanised dialogues

informal chat register

3 Labelling and audit

Automated labelling

Gemini Flash: intent,
outcome and reason

Human audit

all 700 dialogues, line by line

Frozen benchmark

6 to 32 turns, 16.2 on average

Labels used

canonical 262 no / 238 yes
combined 362 no / 338 yes